

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2262

Stepstone 100 kV
October 03, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2262. The Interconnection Customer proposes a 227.6 MW solar photovoltaic facility interconnecting at the Stepstone 100 kV substation in the Indiana area. The request was received on December 24, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 521 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,921,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Segs 7 - Concord Substation 100kV line	Segs 7 Substation - Concord 100kV Line 1	152.8	292	56.2
Segs 7 - Freehaven-Lakepark Substation 100kV Ckt 2	Segs 7 Substation - Freehaven-Lakepark 100kV Line	146.8	322	56.3
Segs 7 - Concord 100kV Ckt 1	Segs 7 - Concord Substation 100kV Line	137.4	324	56.2
Segs 7 - Freehaven-Lakepark 100kV	Segs 7 - Freehaven-Lakepark 100kV	133.1	355	56.3
Twin Branch - Freehaven-Lakepark 100kV	Whitestone - Stepstone 100kV	124.9	392	42.8
Twin Branch - Kemper County 100kV L	Whitestone - Stepstone 100kV line	122.0	427	42.8
Grandview Ave Substation - Drewry Substation 100kV	Table Mt - Gaston Rd 100kV	103.5	281	5.7
Bonnet Crk - Buchanan Mi 100kV	Table Mt - Gaston Rd 100kV	102.6	214	9.4
Buchanan Mi - Northern 100kV L	Table Mt - Gaston Rd 100kV L	102.6	213	9.4
Northern - Ager 100kV	Table Mt - Gaston Rd Substation 100kV	100.9	243	9.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Segs 7 - Concord Substation 100kV line	Rebuild line with higher-capacity conductor	\$3,664,000	26% (\$942,000)
Segs 7 - Freehaven-Lakepark 100kV #2	Rebuild line with higher-capacity conductor	\$2,967,000	26% (\$762,000)
Segs 7 - Concord 100kV Circuit 1	Rebuild line with higher-capacity conductor	\$3,533,000	26% (\$908,000)
Freehaven-Lakepark - Segs 7 100kV Line 1	Rebuild line with higher-capacity conductor	\$2,152,000	26% (\$553,000)
Twin Branch - Freehaven-Lakepark 100kV Line	Rebuild line with higher-capacity conductor	\$2,065,000	22% (\$449,000)
Twin Branch - Kemper County 100kV L	Rebuild line with higher-capacity conductor	\$4,366,000	22% (\$949,000)
Drewry - Grandview Ave 100kV	Reconductor line	\$626,000	4% (\$27,000)
Bonnet Crk - Buchanan Mi 100kV	Reconductor line	\$992,000	9% (\$88,000)
Buchanan Mi - Northern 100kV L	Reconductor line	\$1,446,000	9% (\$128,000)
Northern - Ager 100kV L	Reconductor line	\$1,298,000	9% (\$115,000)
	Total allocated to project		\$4,921,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.